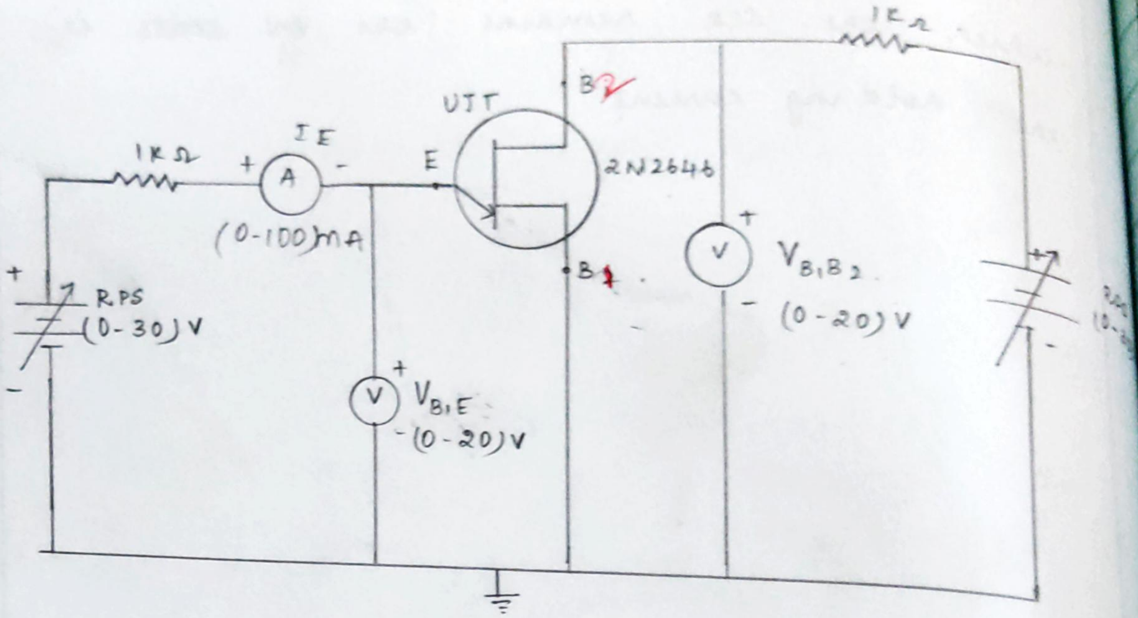
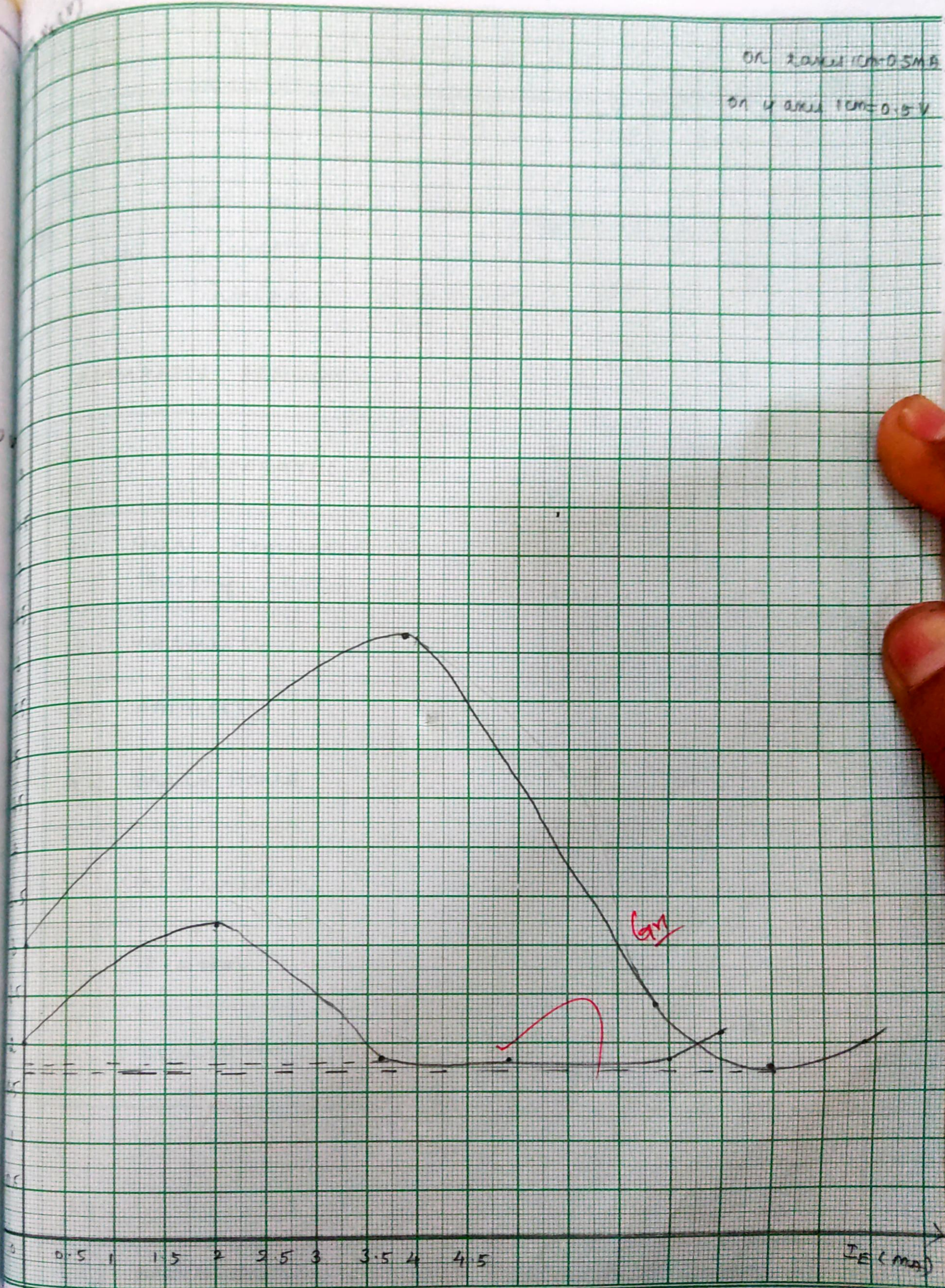
CIRCUIT DIAGRAM :

UJT :



VI characteristics of UJT



$$\eta_1 = \frac{V_p - V_v}{V_{BB}}.$$

$$= 3.29 - 1.47$$

$$\eta_1 = 0.36A.$$

$$\eta_2 = \frac{V_p - V_v}{V_{BB}}.$$

$$= \frac{6.2 - 1.4}{10}$$

$$\eta_2 = 0.48\,A.$$

$$\eta = \frac{\eta_1 + \eta_2}{2}$$

$$= \frac{0.844}{2}$$

$$\eta = 0.42 \text{ (no units)}$$

b) CHARACTERISTICS OF UJT.

AIM:

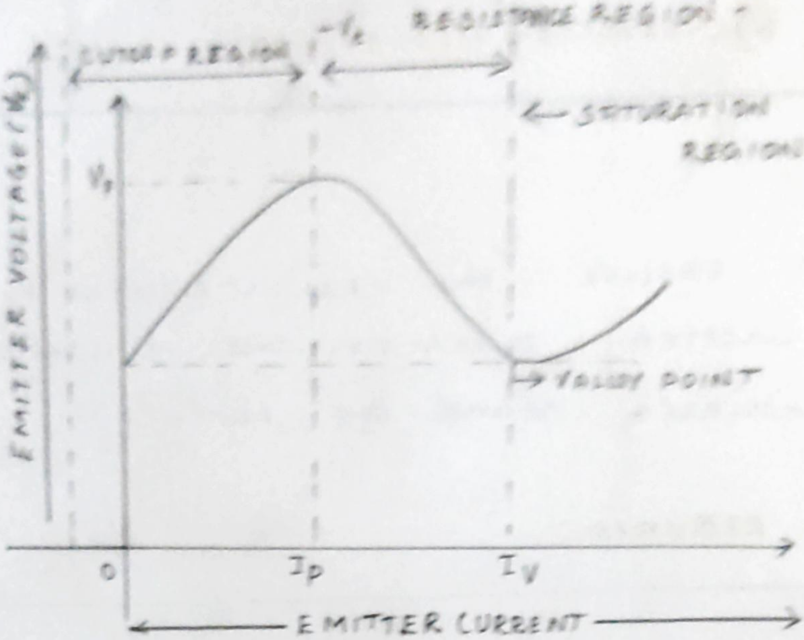
To observe the characteristics of uni-junction transistor and to calculate the intrinsic stand-off ratio (η).

APPARATUS REQUIRED:

SL NO.	APPARATUS NAME	TYPE / RANGE	QUANTITY.
1.	Dual Regulated power supply (RPS)	(0-30) V	1
2.	Voltmeter.	(0-20) V	2
3.	Ammeter	(0-100) mA	1
4.	Bread Board	-	1
5.	Uni Junction Transistor	2N2646	1
6.	Resistor	470 Ω	1
7.	connecting	-	As required.

MODEL GRAPH :

UJT



TABULATIONS :

$V_{BB} = 5V$		$V_{BB} = 10V$	
$V_{EB}(V)$	$I_E(mA)$	$V_{EB}(V)$	$I_E(mA)$
0.5	0	0.5	0
1	0	1	0
1.5	0	2	0
2	0	3	0
2.5	0	4	0
3	0	5	0
3.5	0	6	0
4.5	1.32	6.5	1.33
1.78	1.36	7.2	1.69
1.5	1.35	5.02	1.92
1.4	8.10	4.2	2.88
2.1	8.76	1.91	3.10
		2	6.24

FORMULA :

Intrinsic stand - off ratio (η) = $\dfrac{V_P - V_D}{V_{BB}}$.

V_P - Peak point voltage.

V_D - Diode cut - in - voltage.

V_{BB} - Voltage between base 1 and base 2.

(i.e) Inter - base voltage.

PROCEDURE :

* Connections are made as per the circuit diagram.

* Output voltage is fixed at a constant level and by varying input voltage corresponding emitter current value are noted down.

* This procedure is repeated for different value of output voltages.

* All the readings are tabulated and intrinsic stand-off ratio is calculated using $\eta = \dfrac{V_P - V_D}{V_{BB}}$.

* A graph is plotted between V_{EE} + I_E for different values of V_{BE}

RESULT :

Thus the characteristics of uni-juncti-on transistor was determined and Intrinsic stand-off ratio (η) are determined.

(i.e) Intrinsic stand off ratio (η) = 0.2